

INFECTIOUS DISEASES

An mRNA-based broad-spectrum antiviral inspired by ISG15 deficiency protects against viral infections in vitro and in vivo

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Type I interferons (IFN-I) are cytokines with potent antiviral and inflammatory capacities. IFN-I signaling drives the expression of thousands of IFN-I-stimulated genes (ISGs), whose aggregate function results in the control of viral infections. A few of these ISGs are tasked with negatively regulating the IFN-I response to prevent overt inflammation. ISG15 is a negative regulator whose absence leads to persistent, low-grade elevation of ISG expression and concurrent, often self-resolving, mild autoinflammation. The limited breadth and low-grade persistence of ISGs expressed in ISG15 deficiency are sufficient to confer broad-spectrum antiviral resistance. Inspired by the antiviral state of humans with ISG15 deficiency, we identified a nominal collection of 10 ISGs that recapitulated the broad antiviral potential of the IFN-I system, which typically induces the expression of thousands of ISGs. The expression of this 10-ISG collection in an IFN-I-nonresponsive cell line increased cellular resistance to Zika virus, vesicular stomatitis virus, and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). A lipid nanoparticle-encapsulated messenger RNA (mRNA) formulation of this 10-ISG collection reduced influenza A virus plaque size in samples collected from infected mice when given prophylactically. Moreover, when used collectively and delivered prophylactically, the 10-ISG collection was able to protect hamsters against a lethal SARS-CoV-2 challenge, in contrast with the lack of efficacy when mRNAs were delivered individually. These findings suggest that these 10 ISGs have potential as a broad-spectrum antiviral prophylactic.

INTRODUCTION

Viral pathogens are substantial contributors to worldwide morbidity and mortality. The 1918 influenza pandemic is estimated to have killed 20 to 50 million Europeans (1–3). A hundred years later, despite vaccinations and available therapeutics, it is estimated that influenza virus continues to hospitalize about half a million individuals annually (4). Most recently, the severe acute respiratory syndrome

coronavirus 2 (SARS-CoV-2) pandemic has resulted in 10 to 15 million deaths (5). Given climate change and rapid globalization practices, another viral pandemic is likely imminent (6–8). Therefore, we must develop effective broad-spectrum antivirals to protect us from known, emerging, and unknown viral pathogens (9, 10).

Soon after their initial discovery as antivirals, type I interferons (IFN-I) were thought to be an ideal broad-spectrum therapeutic agent (11–13). Unlike type III IFN (IFN-III), a related cytokine that uses a similar signaling cascade and is showing promise (14), clinical experience has demonstrated limited utility for IFN-I in the treatment and prevention of acute viral infections (15). As a potent inducer of inflammation, IFN-I is poorly tolerated, with substantial side effects (15–17). Recently, the therapeutic efficacy of IFN-I was shown to be disrupted by its induction of various cellular death pathways after the therapy (18). However, modifying the quantity and delivery methods might improve this (19). In addition, IFN-I is self-limiting therapeutically because, among the more than 1000 IFN-stimulated genes (ISGs) that act in concert to restrict viral replication, several negative regulators are present that counteract IFN signaling (20, 21). Although essential in the context of natural infection to limit inflammatory damage to the host, these regulators are a hurdle to IFN-I therapeutic efficacy (21). Their function creates a refractory state that prevents ISG expression in response to sustained stimulation (22, 23).

Ubiquitin-specific peptidase 18 (USP18), in complex with ISG15, plays a prepotent role (24–27). Upon expression, USP18 suppresses IFN-I signaling by physically displacing Janus kinase 1 (JAK1) from

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the IFN- α receptor (IFN α R) complex (28), and ISG15 functions to stabilize USP18, protecting it from proteasome-mediated degradation (29, 30). As a result, deficiency of either ISG15 or USP18 cripples the system's negative regulatory capacity, leading to persistent expression of ISGs and consequences for human health (31). ISG15 deficiency results in partial loss of USP18 activity, leading to a mild form of persistent ISG expression. This has been observed in individuals from more than 10 *ISG15*^{-/-} families, many of whom are well into adulthood (30–33). On the other hand, a complete loss of USP18 results in a much starker clinical phenotype with a severe perinatal onset (31, 34, 35). These two deficiencies thus point to safe and detrimental degrees of IFN-I-mediated inflammation (31, 36).

Underscoring the protective potential of safe degrees of IFN-I-mediated inflammation, our group previously demonstrated that persistent ISG expression in immortalized *ISG15*^{-/-} patient-derived dermal fibroblasts resulted in greater control of a broad spectrum of human pathogenic RNA and DNA viruses (37). In these in vitro experiments, we mimicked the human ISG15-deficient ex vivo phenotype by first stimulating cells with IFN-I and then allowing them to rest (prime rested) before viral infection (37). At this point, in control cells, ISG15 and USP18 have shut down most ISG transcription. However, cells derived from ISG15-deficient individuals with incomplete negative feedback retain expression of a subset of ISGs at ~1% of maximum expression. This is akin to the expression of ISGs detected in the peripheral blood of ISG15-deficient individuals (37). Cells were then virally challenged at a range of multiplicities of infection (MOIs) with a panel of RNA and DNA viruses (37). Across all viral infections tested, fibroblasts derived from ISG15-deficient individuals restricted infection more than healthy control fibroblasts, sometimes by several orders of magnitude (37). Inspired by the ability of ISG15-deficient cells to restrict a broad spectrum of viral infections with only a small fraction and low intensity of the ISG repertoire, we sought to define, formulate, and develop a therapeutic combination of ISGs capable of similar antiviral activity.

RESULTS

ISG15 deficiency inspired the identification of a nominal collection of 10 ISGs with antiviral potential

ISG15^{-/-} individuals live with constant but low ISG expression in their peripheral blood (30). This is due to tonic, environmental, and commensal-induced IFN-I, which these individuals incompletely control. Under sterile culture conditions, *ISG15*^{-/-} cells do not experience baseline ISG expression. To recapitulate the in vivo effect of defective negative feedback in vitro, we stimulated cells with IFN-I for 12 hours, washed them to remove the IFN-I, and incubated them for another 36 hours. By this point, 48 hours after IFN-I exposure, USP18, stabilized by ISG15, down-regulated most ISG transcription in fibroblasts from healthy controls—a process that is incomplete in *ISG15*^{-/-} individuals (Fig. 1A). Because IFN-I primarily induces ISG15 and USP18, their absence does not affect initial IFN-I activation or peak ISG expression. However, ISG15 deficiency results in persistent, low expression of most ISGs, as demonstrated by increases in *IFIT1*, *RSAD2*, and *MX1* mRNA; *BST2* and *SIGLEC1* were not different (Fig. 1A). This residual ISG expression, present at ~1% of peak IFN-I-induced activity, is sufficient to substantially restrict a broad spectrum of viral agents (37). Next, we tested whether the previously documented panviral resistance (37) in *ISG15*^{-/-} human

immortalized fibroblasts extended to Zika virus (ZIKV), given its relevance in past outbreaks. In an initial experiment, overnight ZIKV challenge at an MOI of 1 in IFN-I-exposed fibroblasts suggested a 14-fold reduction of infection in *ISG15*^{-/-} compared with that in similarly treated healthy control cells (Fig. 1B). This apparent inhibition corresponded with decreased productive virus formation (Fig. 1C).

To gain insights into the ISGs responsible for broad-spectrum antiviral activity, we performed RNA sequencing (RNA-seq) on fibroblasts from *ISG15*^{-/-} and healthy individuals under two conditions: unstimulated and IFN-I exposed (12-hour IFN-I stimulation followed by 24-hour rest). *ISG15*^{-/-} fibroblasts harbored either a nonsense or frameshift mutation at position c.379 or c.336_337, respectively. Although these mutations result in total loss of protein expression, their location in the last exon of *ISG15* circumvents RNA decay (30, 32), allowing detection of the transcript by RNA-seq. Difference-of-differences gene expression analyses identified 61 ISGs that were differentially responsive to IFN-I between *ISG15*^{-/-} and control cells (Fig. 1D). In addition, we noticed that a small subset of the differentially regulated ISGs in *ISG15*^{-/-} cells also remained elevated above baseline in the prime-rested fibroblasts from healthy controls, albeit at lower amounts (Fig. 1E). IFN-I-exposed healthy controls retain some antiviral protection (Fig. 1, B and C), suggesting that this antiviral function could be dose dependent on ISG expression. Thus, we hypothesized that the ISGs with lingering expression in healthy controls evolved lower shutdown kinetics because maintaining low-grade expression of a subset of ISGs is sufficient to exert some antiviral effects (Fig. 1, B and C). Therefore, heightened expression of select ISGs may contribute to the improved resistance in IFN-I-exposed *ISG15*^{-/-} cells (Fig. 1E). On the basis of these observations and the antiviral function that each ISG is known to have at distinct steps of the prototypical viral life cycle, we chose 10 persistently elevated ISG effectors (Fig. 1, E and F, and fig. S1). We hypothesized that they were sufficient to replicate the broad-spectrum viral resistance documented in *ISG15* deficiency. This approach is parallel to one taken by highly active anti-retroviral therapy for HIV, which similarly targets various stages of the HIV life cycle (Fig. 1F and fig. S1).

Select persistently elevated ISGs, expressed in isolation, show known but limited antiviral restriction

To test the antiviral potential of these 10 ISGs, we stably transduced each ISG into a *JAK1*^{-/-}, IFN-I-nonresponsive fibrosarcoma (U4C) cell line (Fig. 2A). Leveraging the inability of these cells to respond to IFN-I enabled us to assess the antiviral capacity of each ISG, independent of any endogenous IFN-I produced because of viral challenge (Fig. 2A). After validating ISG expression in their respective cell lines (Fig. 2B), we tested the antiviral potency of these ISGs against a panel of RNA viruses: vesicular stomatitis virus (VSV), influenza A virus (IAV H5N1), and ZIKV (PRVABC59). As expected, myxovirus resistance 1 (MX1) was sufficient to restrict VSV and IAV H5N1 infection (Fig. 2, C and D), whereas interferon alpha-inducible protein 6 (IFI6) proved capable against ZIKV (Fig. 2E). Although IFI6 showed antiviral activity against ZIKV, infection was further controlled by IFN-I-exposed fibroblasts from *ISG15*^{-/-} individuals (Figs. 1, B and C, and 2E). Observing that IFN-I-exposed *ISG15*^{-/-} fibroblasts outperformed the established panflavivirus restriction factor IFI6 in a ZIKV challenge led us to postulate that the combination of the 10 ISGs may work synergistically to restrict viral infection.

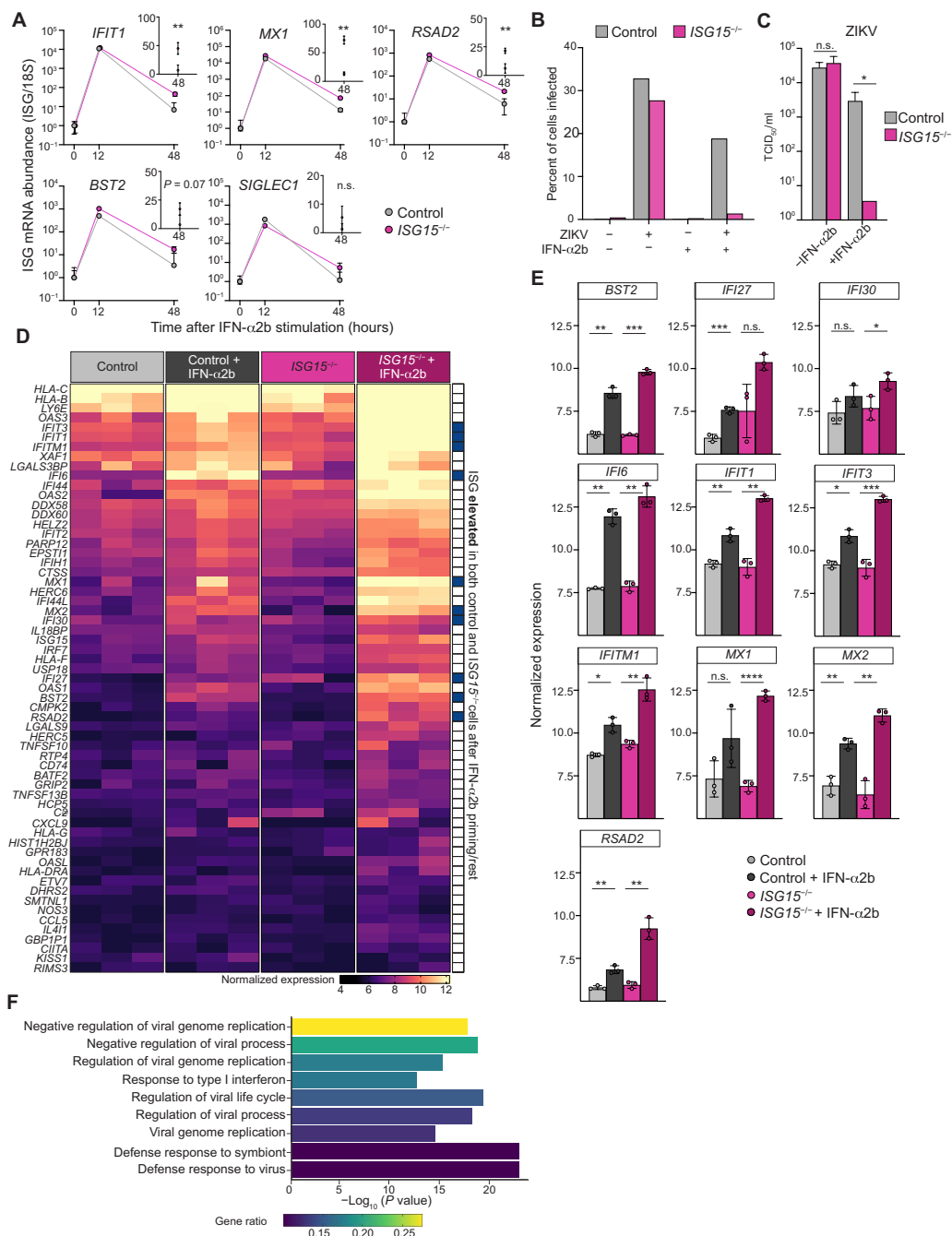


Fig. 1. ISG15 deficiency inspires the identification of a nominal collection of 10 ISGs with broad-spectrum antiviral potential. (A) Relative mRNA time-course quantification of *IFIT1*, *BST2*, *RSAD2*, *MX1*, and *SIGLEC1* from *ISG15*^{-/-} individuals (n = 3) or control (n = 3) hTERT-immortalized fibroblasts treated with IFN-α2b (1000 IU/ml) for 12 hours, washed, and allowed to rest for 36 hours posttreatment. Not significant (n.s.); **P < 0.01, using paired, two-sided Welch's t tests. Data are shown as mean ± SD (n = 3 biological replicates). Missing error bars indicate either very small SDs or that the lower error bars extend below zero, which could not be displayed on a logarithmic scale. (B) After the 48-hour combined IFN-α2b stimulation and resting period, cells were infected with ZIKV (PRVABC59) at an MOI of 1.0 overnight, fixed, subjected to live/dead and viral staining for ZIKV envelope protein (4G2 antibody), and processed and analyzed by flow cytometry. The percentage of infected cells was quantified. (C) Viral titers in supernatants from ZIKV-infected cells were quantified by TCID₅₀ assay using control fibroblasts. n.s.; *P < 0.05 by two-sided paired t test with Welch's correction. Values were log transformed before analysis, and data represent mean ± SD (n = 3 biological replicates). (D) Differentially expressed IFN-I-responsive genes between unstimulated and 12-hour IFN-α2b-stimulated, 24-hour rested fibroblasts from control and *ISG15*^{-/-} individuals. Heatmap color intensity depicts variance-stabilized, normalized expression values. The row annotation bar describes overlap of elevated ISGs in both *ISG15*^{-/-} and control IFN-α2b-exposed conditions (relative to unstimulated; genotype matched). (E) Normalized expression values of 10 selected ISGs elevated in both *ISG15*^{-/-} and control IFN-α2b-exposed conditions, compared with respective unstimulated controls. *IFI30* and *MX1* did not meet the significance cutoff but were included because of their antiviral functions. *IFIH1* was excluded because of its role as a double-stranded RNA sensor and signal potentiator of the IFN-I response. n.s.; *P_{adj} < 0.05, **P_{adj} < 0.01, ***P_{adj} < 0.001, and ****P_{adj} < 0.0001 by Benjamin-Hochberg method. (F) GO enrichment analysis of differentially IFN-I-responsive genes indicated in (D). Gene ratio indicates the proportion of genes in the input list that are associated with each GO term compared with the total number of genes shown in (D).

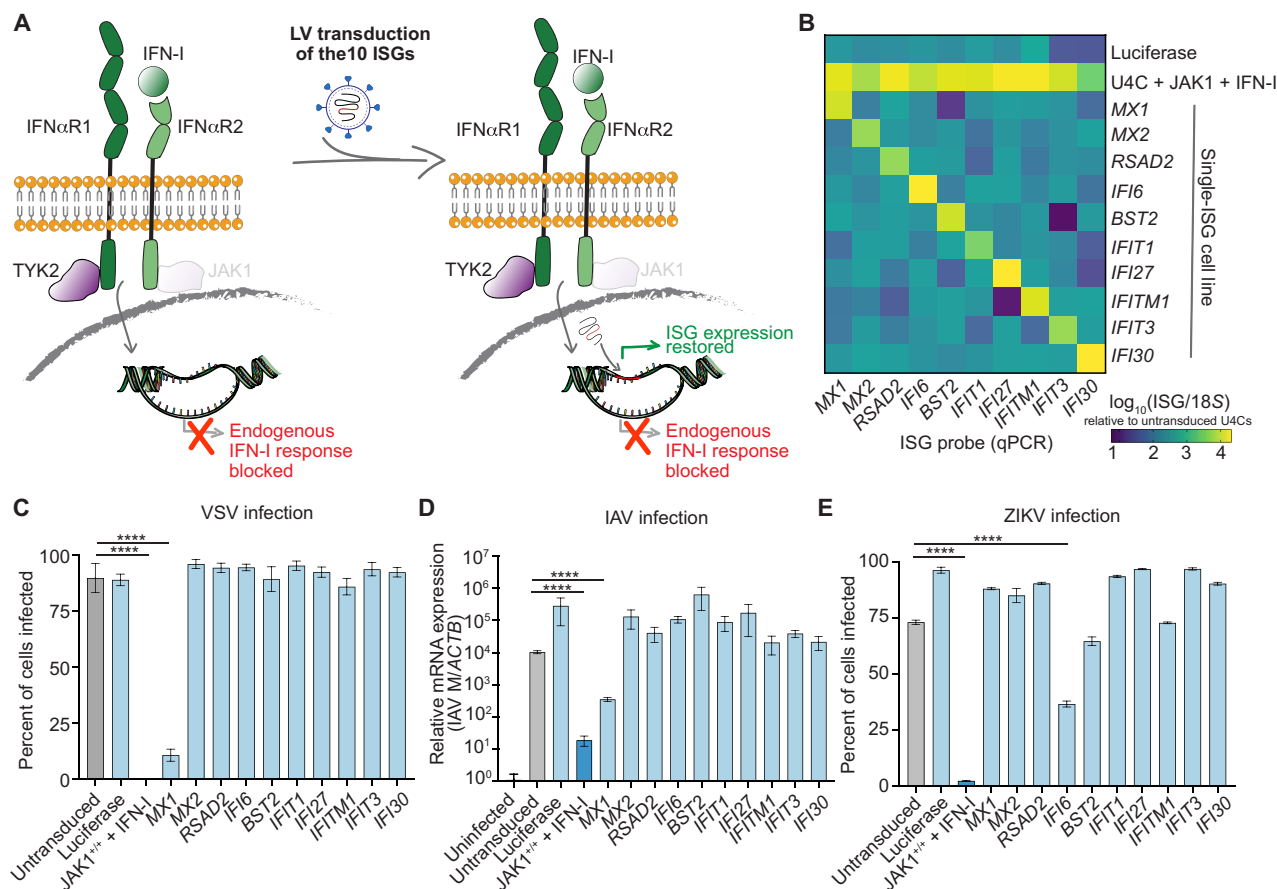


Fig. 2. Select persistently elevated ISGs expressed in isolation show known but limited antiviral restriction. (A) Schematic of in vitro $JAK1^{-/-}$ IFN-I-nonresponsive cell system to study direct ISG antiviral effects. Cells were transduced with lentiviruses encoding an individual ISG (from 10-ISG collection) or luciferase before viral challenge. TYK2, tyrosine kinase 2; LV, lentiviral vector. (B) Relative qPCR quantification of mRNA transcripts of encoded ISGs in untransduced and lentivirus-transduced lines. (C to E) Lentivirus-transduced cells were subjected to viral challenge with VSV [MOI: 2; 8-hour infection (C)], H5N1 Vietnam halo IAV [MOI: 2; 1-hour infection (D)], or ZIKV [MOI: approximately 2, 24-hour infection (E)]; the percentage of infected cells was determined by microscopy [(C) and (E)], and viral mRNA was quantified by RT-qPCR (D). Quantification of IAV matrix gene (IAV M) mRNA transcripts in (D) is shown relative to uninfected controls. **** $P < 0.0001$ by one-way ANOVA with Dunnett's test, comparing each group with the untransduced control. Only groups with significantly reduced infection are marked. Data are shown as mean $\pm$ SD ($n = 3$ biological replicates).

Synthetically modified RNA delivery of 10-ISG collection is sufficient to restrict viral infection in vitro

To interrogate this, we adopted a synthetic, modified RNA (modRNA) technology coupled with lipid nanoparticle (LNP) delivery to introduce a collection of 10 ISGs (modISG10) (table S1) into cells before viral challenge (Fig. 3A). Using a modRNA-LNP encoding emerald green fluorescent protein (GFP), we detected high transfection efficiency in U4C $JAK1^{-/-}$ IFN-I-nonresponsive cells and confirmed that modRNA encoding GFP is translatable (Fig. 3B).

To assess the efficiency and immunogenicity of 10-ISG modRNA-LNP transfection, we measured the abundance of ISG-encoding modRNA 16 hours posttransfection (Fig. 3C). We detected mRNA of all 10 ISGs, and the extent of ectopic modRNA ISGs detected was comparable to, and in some cases greater than, the peak expression induced by IFN-I stimulation (Fig. 3C). Expression of GFP and all 10 ISGs at the protein level was also confirmed using either mass spectrometry (MS) or immunoblotting (Fig. 3, D and E, and fig. S2). In addition, to demonstrate the specificity of 10-ISG modRNA-LNP delivery, we confirmed that the expression of *USP18*, an ISG not included in the 10-ISG combination, remained unchanged compared

to that in the untreated and luciferase-transfected conditions (Fig. 3C). Similarly, delivery of modRNA-LNP encoding mouse ISG orthologs into human-derived human embryonic kidney (HEK) 293T cells did not result in the induction of human *USP18*, *ISG15*, and *IFI44L*, ISGs unrelated to the delivered subset, which were potentially induced by IFN-I stimulation (fig. S3, A to C, and table S2). Consistently, induction of IFN-I, IFN-II, or IFN-III was not observed in U4C, A549, or 2fTGH cells transfected with the 10 ISGs (fig. S3, D to F). In parallel, we tested whether delivery of the 10 ISGs drives a refractory state that halts subsequent activation of the IFN-I system. A549 cells were mock-treated, transfected with the 10 ISGs, or stimulated with IFN-I and rested before a second stimulation with IFN-I. As expected, we found that cells pretreated with IFN-I entered a refractory state, evidenced by the accumulation of *USP18* and down-regulation of phosphorylated signal transducer and activator of transcription 1 (pSTAT1) and pSTAT2 (fig. S4). In contrast, cells primed with modISG10 responded to the second IFN-I stimulation with a dose-dependent induction of pSTAT1 and pSTAT2, resembling mock-treated cells in their *USP18* abundance. Together, these observations suggested that the addition of modRNAs to cells was

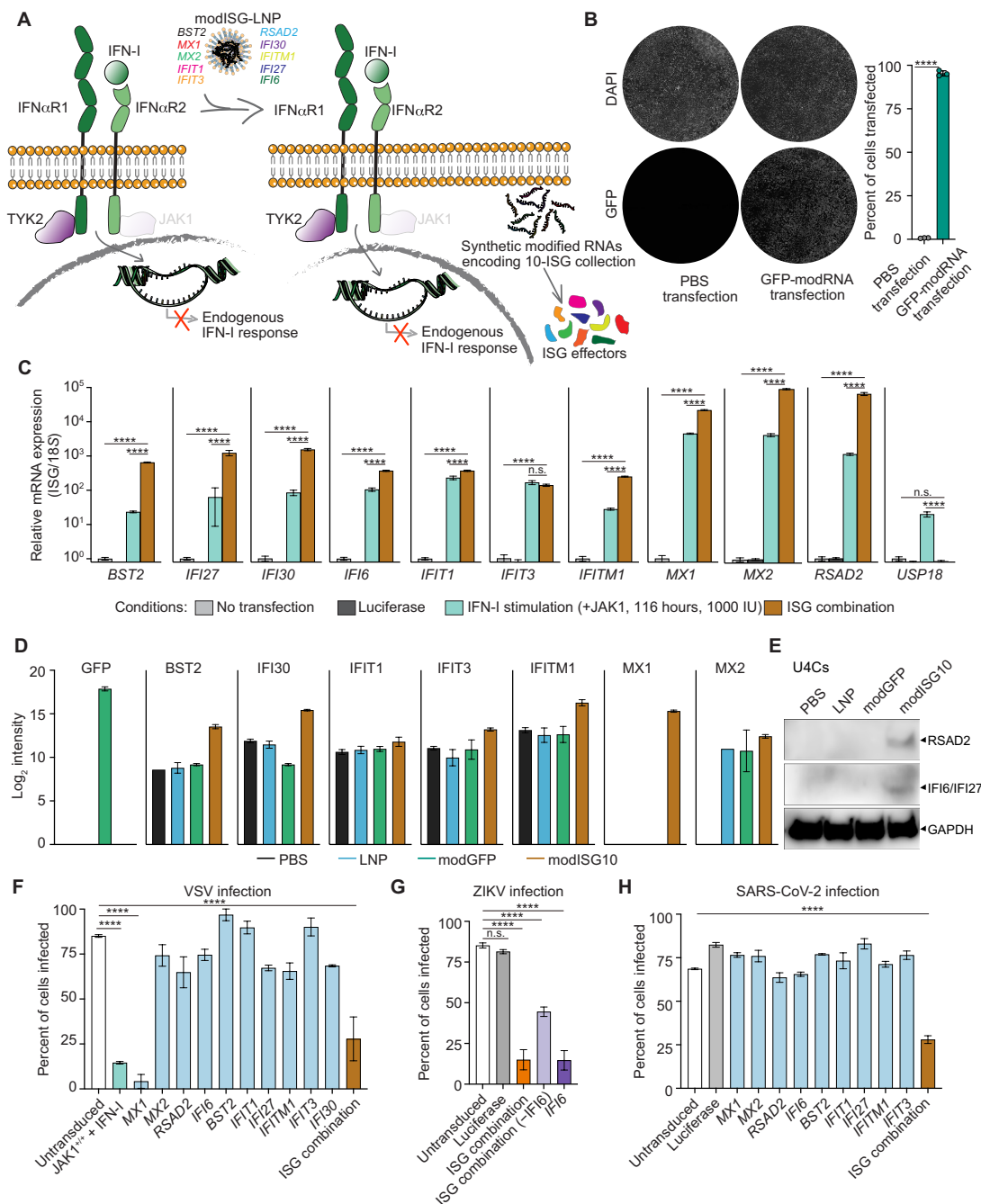


Fig. 3. Synthetic modRNA delivery of 10-ISG collection is sufficient to restrict viral infection. (A) Schematic of synthetic modRNA delivery in the IFN-I *JAK1*^{-/-} nonresponsive cell system. ModRNAs encoding emerald GFP or the 10-ISG collection were pooled in an equimolar ratio, packaged into LNPs, and transfected into cells for 16 hours before imaging, relative mRNA quantification, and viral challenge. (B) Representative bright-field and GFP images of cells treated with PBS or modRNA-LNP encoding emerald GFP, 16 hours posttransfection (left). Quantification of GFP⁺ cells (right). (C) Relative qPCR quantification of exogenously expressed 10 ISGs and *USP18* in nontransfected, luciferase modRNA-LNP-transfected, JAK1-complemented and IFN-I-prestimulated, and 10-ISG collection modRNA-LNP-transfected cells at 16 hours posttransfection. *USP18* expression was measured to assess nonspecific IFN activity of modRNA-LNP administration. n.s.; *****P* < 0.0001 by one-way ANOVA with Tukey's test. Data are shown as mean ± SD (*n* = 3 biological replicates). (D) Abundance of emerald GFP and seven different ISGs (BST2, IFI30, IFIT1, IFIT3, IFITM1, MX1, and MX2) was quantified in nontransfected U4Cs or U4Cs transfected with RNAiMAX Lipofectamine alone (LNP), modGFP RNA, or modISG10 RNA, as assessed by MS. Bars represent the mean ± SD (*n* = 3 biological replicates) of log₂-transformed intensity values for each ISG indicated. Error bars are shown only for proteins detected in at least two of the three samples within a group. (E) RSAD2 and IFI6/IFI27 proteins were detected by immunoblot of U4Cs transfected with modRNA of 10-ISG collection. GAPDH was used as the loading control. (F and H) ModRNA-LNP-transfected cells were subjected to viral challenge with VSV [MOI: 2; 8-hour infection (F)], ZIKV [MOI: approximately 2; 24-hour infection (G)], or SARS-CoV-2 virus [MOI: ~2.5; 24-hour infection (H)]; the percentage of infected cells was determined by microscopy. n.s.; *****P* < 0.0001 by one-way ANOVA with Dunnett's test, comparing each group with the untransduced control. Data are shown as mean ± SD (*n* = 3 biological replicates). For SARS-CoV-2 infections, U4C *JAK1*^{-/-} IFN-I-nonresponsive cells were transduced with human ACE2 to make cells susceptible to viral infection. Bars represent average infection values between three biological replicates.

introducing the effector function of the encoded ISGs without the side effects of IFN-I.

Next, to assess whether modRNA-LNP delivery of ISGs was sufficient to restrict viral infection, we took cells transfected with single ISGs or the 10-ISG collection and challenged them with VSV (Fig. 3F). As expected, and as previously demonstrated, cells transfected with *MX1* modRNA restricted VSV infection similar to *JAK1*-complemented cells prestimulated with IFN-I. No other single ISG appeared to restrict infection relative to the nontransfected control. However, transfection with the 10-ISG collection controlled infection to similar degrees as *JAK1*-expressing cells prestimulated with IFN-I (Fig. 3F). To assess the combinatorial antiviral potential of the 10-ISG collection, we revisited our ZIKV infection model. Considering our observation of a moderate restriction of ZIKV by IFI6 (Fig. 2E) and the modular nature of the modRNA-LNP cocktail generation process, we added an additional transfection condition to deliver nine ISGs (excluding *IFI6*). As expected, the 10-ISG combination and IFI6 alone restricted ZIKV infection (Fig. 3G). The nine-ISG combination cocktail, in the absence of IFI6, also appreciably restricted ZIKV infection, suggesting that the other ISG effectors act in concert by unknown mechanisms to stem infection (Fig. 3G). In addition, we tested the efficacy of the 10-ISG collection against another flavivirus, West Nile virus (WNV). We found that pretreatment with modISG10, but not modIFI6 (modRNA encoding IFI6) or modGFP (modRNA encoding GFP), protected 2fTGH cells from WNV infection (fig. S5A). Similarly, pretreatment with modISG10 protected B6 mouse embryonic fibroblasts from WNV infection similar to IFN-I (fig. S5B).

Next, we further explored the antiviral capacity of our 10-ISG collection to restrict SARS-CoV-2 infection (Fig. 3H). Single-ISG modRNA-LNP transfections had minimal to no effect on restricting viral infection of IFN-I–nonresponsive *JAK1*^{−/−} fibrosarcoma cells. However, an antiviral effect was observed when the 10 ISGs were expressed in combination (Fig. 3H). To test whether the antiviral effect observed after delivery of the 10 ISGs was accompanied by cell death, we transfected fibroblasts with LNP alone, modGFP, modIFI6, or modISG10 or treated them with IFN-I. There were no detectable differences in cell viability among all conditions at 24 and 48 hours after treatment (fig. S6, A and B). The total percentage of live cells across the tested conditions was comparable to the percentage of cells left in medium alone (fig. S6, A and B). Only fibroblasts incubated with cell death–inducing agents shikonin and 10% dimethyl sulfoxide (DMSO) showed a decline in cell viability. Combined, these data suggest that the 10-modRNA combination restricts VSV, IAV, ZIKV, WNV, and SARS-CoV-2 infection in vitro.

Synthetic modRNA delivery of the 10-ISG collection is sufficient to restrict IAV and SARS-CoV-2 viral load in vivo

Having validated the potency of the 10-ISG collection against multiple viruses in vitro, we next sought to assess its ability to restrict infection in vivo. We first tested the antiviral efficacy of the 10-ISG collection in an in vivo mouse model of IAV PR8 infection. Achieving efficient delivery of modRNA-LNP to mouse lungs was technically challenging. As a result, only modest expression of emerald GFP was detected in the lungs 24 hours after modGFP administration (fig. S7, A to C). Despite this, we evaluated the antiviral potency of the 10-ISG syndicate. Mice were treated on day −1 with phosphate-buffered saline (PBS), IFN- α 2b, modRNA-LNP encoding mouse ISG orthologs (fig. S8 and table S2), or emerald GFP and then challenged

with a lethal dose ($5\times$ LD₅₀) of IAV PR8 on day 0 (fig. S9A). Mice were monitored for viral titers at day 3 after infection, body weight loss, and survival. Among the treatment groups, only modMX1 (modRNA encoding MX1) reduced viral titers in the lungs compared with PBS. In contrast, plaque size measurements showed that both modMX1 and modISG10 reduced plaque size relative to PBS, with a more modest reduction observed in the IFN- α 2b–treated group (fig. S9, B and C). No effect was observed in the modGFP group. However, in the context of this lethal challenge, neither modMX1 nor modISG10 treatment improved survival or prevented body weight loss (fig. S9, D and E), likely because of low lung modRNA expression in this system. To determine whether the delivery of packaged modISG10 RNA induces IFN production in vivo, we measured serum concentrations of IFN- β , IFN- γ , and IFN- λ 2/3 in IAV-infected mock-, modRNA-LNP-, or IFN-I–treated mice. We did not detect any differences in serum IFN- β , IFN- γ , and IFN- λ 2/3 among all the groups (fig. S10, A to C). Although prophylactic treatment of mice with the 10-ISG collection did not rescue survival after a lethal viral challenge, even the limited expression achieved in this system was sufficient to reduce IAV PR8 replication in the lung.

Next, we adopted the Syrian golden hamster (*Mesocricetus auratus*) model to evaluate SARS-CoV-2 infection (Fig. 4A). We first assessed the efficiency of two different administration routes for delivering modRNA-LNPs to hamster lungs: intranasal and intratracheal (Fig. 4B). Harvesting lung tissue revealed appreciable emerald GFP expression in the lungs of animals from both delivery groups (Fig. 4, B and C), unlike in mice, where expression was more modest (fig. S7, A to C). Next, to assess the antiviral capacity of the 10-ISG combination delivered intranasally, we treated animals with PBS, modRNA-LNP encoding hamster ISG orthologs (fig. S8 and table S3), or emerald GFP 1 day before SARS-CoV-2 viral challenge. Animal weights were monitored, and select tissues were collected at days 2 and 4 postinfection (Fig. 4A). Initially, all animals gained weight; however, starting at day 2 postinfection, differences in weight began to emerge between experimental groups (Fig. 4D). Beginning at day 4 postinfection, compared with PBS-treated animals or animals treated with modRNA-LNP encoding emerald GFP, animals treated with modRNA-LNP encoding the 10 ISGs were protected from sickness-associated weight loss (Fig. 4D). Furthermore, viral titer measurements in the lungs at day 2 postinfection revealed a reduction in the amounts of infectious SARS-CoV-2 virions in the lungs of the 10-ISG combination treatment group compared with those in the lungs of the emerald GFP group (Fig. 4E). Similar differences in viral burden were not detected in the olfactory bulb at the same time point, likely because of the restricted delivery of modISGs to the lungs. Furthermore, in transcriptomic analyses of day 4 SARS-CoV-2–infected hamster lungs, we detected viral transcripts in mock-treated and emerald GFP–expressing control animals. In contrast, detection of viral transcripts was lower in the animals treated with the 10-ISG collection (Fig. 4F). This was accompanied by a difference in host gene expression profiles between animals treated with the 10-ISG collection and control groups (Fig. 4, F and G). Principal components analysis (PCA) of the infected hamster lungs revealed that PBS- and GFP-modRNA–treated samples grouped closely together, whereas they were distinctly separate from the ISG-10–modRNA samples (Fig. 4G). Clinical scoring of infected hamsters' lungs, after hematoxylin and eosin staining on day 7 postinfection, revealed extensive disease pathology as indicated by airway constriction (Fig. 4H). Half of the animals assigned to the

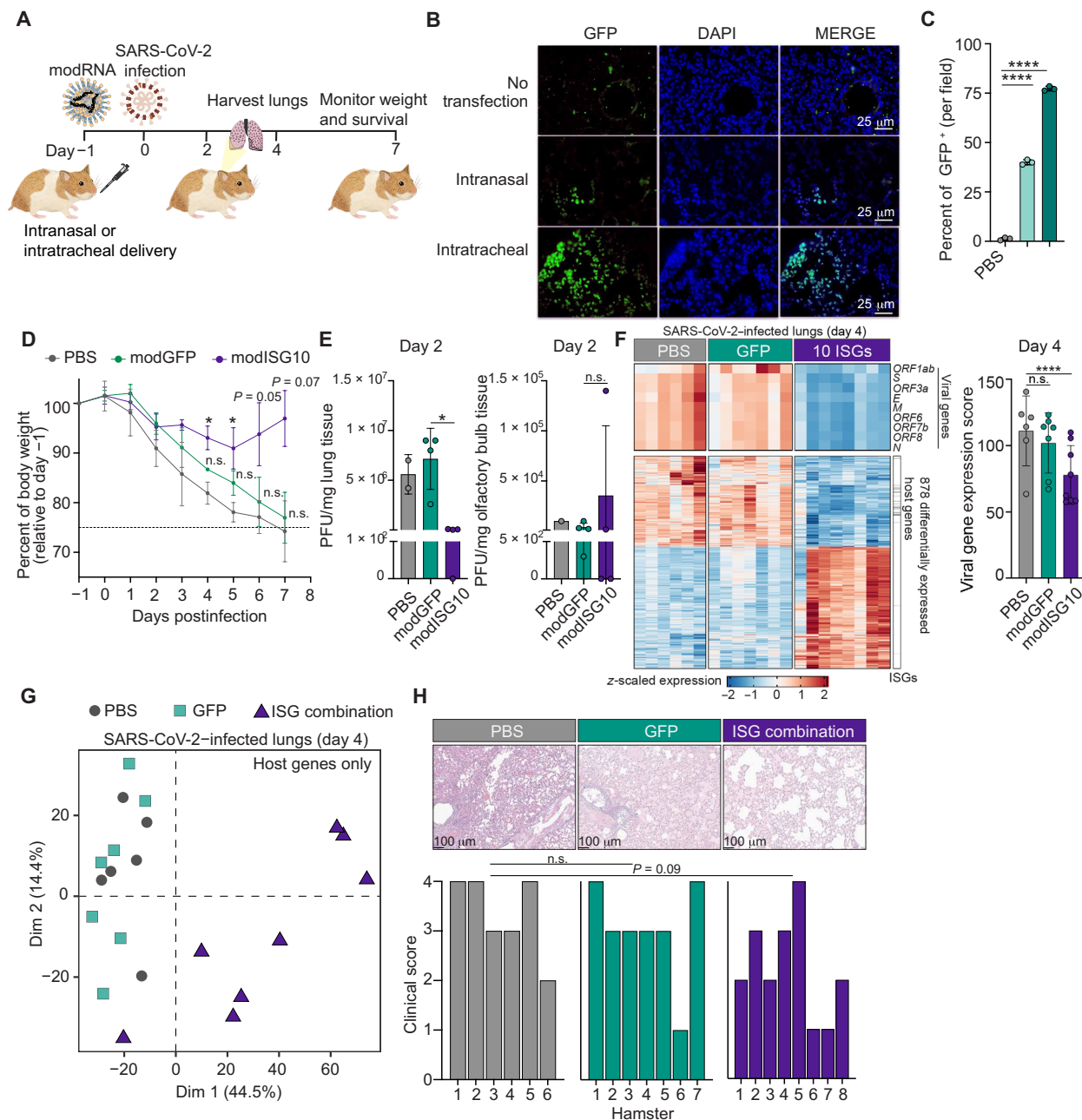


Fig. 4. Intranasal delivery of synthetically modified RNA encoding 10-ISG collection is sufficient to restrict SARS-CoV-2 in vivo. (A) Experimental design of intranasal modRNA-LNP delivery and SARS-CoV-2 viral challenge in the Syrian golden hamster model. (B) Representative confocal microscopy images of day 2 lung sections from emerald GFP modRNA-LNP-treated or PBS-treated hamsters. Scale bars, 25 μ m. (C) Quantification of (B) showing GFP⁺ cells per field. **** P < 0.0001 by one-way ANOVA with Dunnett's test. Data are shown as mean $\pm$ SD (n = 3 animals per group). (D to H) Shown are outcomes after intranasal modRNA-LNP delivery. (D) Weight loss curves over a 7-day infection time course of PBS-treated, emerald GFP modRNA-LNP-treated, and 10-ISG modRNA-LNP-treated hamsters after SARS-CoV-2 infection. The dotted line represents the weight loss threshold per the predefined humane end point. n.s.; * P < 0.05 by mixed-effects model with Geisser-Greenhouse correction and Dunnett's post hoc versus PBS control. Data are shown as mean $\pm$ SD (n = 5 to 8 animals per group). (E) Day 2 SARS-CoV-2 viral loads were measured in PBS- or modRNA-treated hamster lungs (left) and olfactory bulbs (right). n.s.; * P < 0.05 by two-tailed Mann-Whitney test. The PBS groups were excluded from statistical analysis because of insufficient sample size resulting from viremia-related mortality. Data are shown as mean $\pm$ SD (n = 2 to 4 animals per group). (F) Viral gene expression and differentially expressed host genes on day 4 postinfection. Heatmaps show viral gene expression (left) and 878 differentially expressed host genes (right). Composite scores of viral gene expression, based on normalized counts of all viral transcripts detected, are shown. n.s.; **** P < 0.001 by one-way ANOVA with Dunnett's test. Data are shown as mean $\pm$ SD (n = 6 to 8 animals per group). (G) PCA on host gene expression showing separation of 10-ISG modRNA-LNP-treated hamsters from PBS-treated and emerald GFP modRNA-LNP-treated hamster groups (n = 6 to 8 animals per group). Dim, dimension. (H) Representative images (top) and histological scoring (bottom) of PBS- or modRNA-LNP-treated hamster lungs at day 7 postinfection (n = 6 to 8 animals per group). Scale bars, 100 μ m. n.s.; data were analyzed by one-way ANOVA with Dunnett's test. Data are shown as mean $\pm$ SD (n = 6 to 8 animals per group).

modISG10 group exhibited pathology scores of only 1 or 2, suggesting that modRNA-LNP administration of the 10-ISG collection contributes to restricting SARS-CoV-2 infection and mitigating lung inflammation and damage.

DISCUSSION

None of the 67 antiviral drugs currently approved by the US Food and Drug Administration are clinically used as an effective broad-spectrum treatment for viral infections (38). Some viral infections are self-resolving, but many cause severe morbidity and mortality. In 2014, Ebola virus killed 11,000 people in West Africa (39). Since 2015, infections by ZIKV in the Americas have resulted in thousands of infants born with microcephaly (40). Just under 100 years ago, the 1918 influenza pandemic was estimated to have killed 20 to 50 million Europeans (3). Even today, influenza kills 3000 to 50,000 individuals annually in the US alone, depending on the flu strain, with older adults and children most affected (3, 41, 42). During the 2024–2025 influenza season, the US Centers for Disease Control and Prevention (CDC) estimated at least 24 million flu illnesses, 310,000 hospitalizations, and 13,000 deaths so far (43). Recently, an unprecedented pandemic caused by SARS-CoV-2 claimed millions of lives within the past few years (44). History shows that every 2 to 3 years, there is an epidemic of a highly contagious, often deadly, viral disease with tremendous associated morbidities (45, 46). As a society, we are not prepared for future outbreaks of either previously unidentified strains of existing viruses (e.g., SARS-CoV-2 or a repeat of the 1918 flu) or an unknown highly pathogenic virus. The availability of a broad-spectrum antiviral drug would meaningfully increase our ability to fight off known and unknown viral diseases alike. A drug targeting influenza could save 3000 to 50,000 lives annually in the US, and a drug effective against ZIKV could prevent severe morbidities. In addition, an effective drug against SARS-CoV-2 would have saved millions of lives. The development of such drugs would be a substantial leap forward, well worth the effort in terms of our pandemic preparedness.

Genetic studies in humans with inborn errors in the IFN-I system have manifested how indispensable this system is for fighting viral infections (31, 47). In contrast, examples of overt IFN-I signaling due to genetic defects in controlling production or response to IFN-I also define how detrimental prolonged, excessive IFN-I signaling can be (47–50). *ISG15*^{−/−} individuals have likely defined the optimal “Goldilocks zone” of safe amounts of IFN-I signaling (31): sufficient for enhanced control of every virus tested, with minimal side effects (isolated, self-resolving, early childhood skin lesion and Bacille Calmette-Guérin vaccine lymphadenopathy). Notably, most of these individuals are well into adulthood (31). Here, inspired by these individuals’ impressive ability to restrict a broad spectrum of viral infections with only a small fraction of the ISG arsenal, we defined a nominal but sufficient syndicate of ISGs that can achieve similar antiviral activity.

In line with the need for a broad-spectrum antiviral, we leveraged recent advances in RNA therapeutics and LNP technology to deliver synthetic modRNAs expressing the ISG combination in vitro and in vivo. Although RNA delivery remains challenging in general, we show that the ISG combination not only was low in immunogenicity, highly expressed, and well tolerated both in vitro and in vivo but also restricted SARS-CoV-2 infection in hamsters through a synergistic mechanism because each ISG alone was insufficient

to exert control. Ultimately, we engineered a system in which we reduced the ISG repertoire to a 10-ISG effector syndicate that effectively controls viral infection and severity of viral disease. With this, we aimed to extract the essence of antiviral effects while reducing inflammation by omitting the thousands of naturally induced ISGs. Hence, the modality of transient gene therapy, where inhalation of a collection of ISG modRNAs is used, is poised to serve as a true broad-spectrum antiviral against both existing and yet unknown respiratory viruses. With further development, one could envision an à la carte selection of one to three select ISGs for specific viral infections.

Despite the promise of the 10-ISG modRNA-LNP platform, several limitations remain. First, although the approach reduced viral loads and protected hamsters from sickness-associated weight loss, its overall therapeutic impact in mice was constrained by inefficient lung delivery, as evidenced by modest transgene expression and no improvement in survival. Optimizing RNA delivery remains a critical challenge. Second, the antiviral effect of the ISG cocktail appears to be synergistic, but the mechanistic basis for this synergy is not fully defined. Without understanding which ISGs are essential or redundant, rational refinement of the cocktail remains limited. Determining which combinations of the 10 ISGs confer sufficient antiviral effect would require testing more than 1020 combinations for each virus, MOI, and time point, posing a substantial experimental challenge. Third, although the strategy shows potential as a prophylactic, the durability of protection is unclear. Although in vivo stability of delivered mRNA has been reported to last 4 to 7 days, this duration can vary depending on the target organ, delivery efficiency, and administration route (51–53). How far in advance it must be administered and whether it retains efficacy when delivered postinfection remain open questions.

This work lays the foundation for a first-line, deployable antiviral platform that is virus agnostic and modular. Because mRNA delivery technologies continue to advance, future efforts should focus on defining minimal ISG combinations that retain broad-spectrum efficacy with simplified composition. Equally important will be optimizing RNA delivery to ensure efficient, tissue-targeted expression with minimal immunogenicity. Expanded testing in diverse preclinical models, including models of delayed treatment, will be essential to evaluate both prophylactic and therapeutic potentials. Ultimately, this platform could be developed into a versatile antiviral countermeasure, adaptable for rapid deployment against a wide range of viral threats.

MATERIALS AND METHODS

Study design

The objective of this study was to develop and evaluate a synthetic modified mRNA-based delivery platform encoding a rationally curated 10-ISG combination with broad-spectrum antiviral potential. We tested the antiviral efficacy of this 10-ISG modRNA-LNP platform in vitro using IFN-I nonresponsive *JAK1*^{−/−} U4C fibrosarcoma cells and in vivo using murine and Syrian hamster models of IAV and SARS-CoV-2 infection, respectively. Key end points included viral titers determined by half-maximal tissue culture infectious dose (TCID₅₀) or plaque assays, quantification of host and viral gene expression by reverse transcription quantitative polymerase chain reaction (RT-qPCR) or RNA-seq, protein expression analysis by MS or immunoblotting, clinical monitoring of weight loss, and histopathological scoring of lung tissue.

No formal power analysis was performed. Sample size for in vitro experiments (typically $n = 3$ biological replicates) followed standard virological assay design to detect biologically meaningful effects. Group sizes for in vivo experiments ranged from 2 to 10 animals, determined by logistical feasibility and prior experience with detectable effect sizes in antiviral studies. Predefined humane end points were used to stop data collection in animals that lost more than 25% of their baseline body weight or exhibited severe clinical symptoms. All data were included unless there was a technical error in transfection, viral challenge, or measurement, in which case the sample was excluded. Outliers were not excluded unless they violated predefined assay performance criteria.

In vivo treatments (PBS and modRNA-LNP encoding GFP or ISGs) were randomly assigned to animals or cages within experimental batches. In vitro samples were plated and treated in randomized well positions to minimize spatial bias during infection and imaging. For animal experiments, personnel conducting the viral infection and administering treatments were distinct from those evaluating clinical scores for lung histology. In vitro quantification of infection was automated and conducted blind to treatment condition.

Each in vitro experiment was performed with at least three biological replicates, and when samples permitted, at least two independent experiments were conducted to confirm reproducibility. For in vivo studies, the number of biological replicates per group varied and is indicated in the corresponding figure legends; experiments were independently repeated where feasible. For all quantitative assays, technical replicates were averaged per biological replicate before analysis. The study adheres to key principles of the Animal Research: Reporting of In Vivo Experiments (ARRIVE) 2.0 guidelines for the design, conduct, and reporting of animal research, as outlined by the EQUATOR Network.

Cell culture

The Icahn School of Medicine Institutional Review Board approved the use of human cell lines. Dermal fibroblasts from *ISG15*^{-/-} individuals ($n = 3$) and wild-type controls ($n = 3$) were immortalized by stable transduction with human telomerase reverse transcriptase. *ISG15*^{-/-} fibroblasts harbor either a nonsense or frameshift mutation at position c.379 or c.336_337, respectively. Because mutations are in the last exon of *ISG15*, these transcripts are not subject to RNA decay but result in total loss of protein expression (30, 32). HEK293T cells [CRL-11268, American Type Culture Collection (ATCC)], Vero-E6 cells (CRL-1586, ATCC), Vero cells (CCL81, ATCC), U4C *JAK1*^{-/-} cells (a gift from S. Pelligrini), 2fTGH cells (*JAK1*^{+/+} U4C parental; a gift from S. Pelligrini), A549 cells (CCL-185, ATCC), and human fibroblasts were cultured in normal growth medium consisting of Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (FBS), 1% L-glutamine, and 1% penicillin/streptomycin. All cells were cultured at 37°C and 10% CO₂. All cells were tested for mycoplasma contamination using the Plasmotest kit according to the manufacturer's instructions (InvivoGen).

Animals

Six- to eight-week-old female C57BL/6 mice (strain no: 000664) were purchased from the Jackson Laboratory. Four- to six-week-old female Syrian golden hamsters [Crl:LVG(SYR); strain code: 049] were purchased from Charles River Laboratories. The animals were maintained under a strict 12-hour light cycle and fed a standard

chow diet ad libitum in a specific pathogen-free barrier facility at the Icahn School of Medicine at Mount Sinai (ISMMS). All experiments involving animals were performed in accordance with the regulatory guidelines and standards set by the Institutional Animal Care and Use Committee of ISMMS.

Lentivirus generation and relevant ISG-expressing cell lines

Single-ISG open reading frames (ORFs) encoding human *MX1*, *MX2*, *IFIT1*, *IFIT3*, *IFITM1*, *IFI6*, *IFI27*, *IFI30*, *BST2*, and *RSAD2* (table S1) were independently cloned from an IFN- α 2b-stimulated control fibroblast line and inserted into the pTRIP vector backbone by gateway ligation, described previously (54). To generate lentiviruses, 2.5 μ g of the desired transfer vector, 2 μ g of psPAX2, 0.8 μ g of pMD2.G, 14 μ l of Lipofectamine 3000 (Thermo Fisher Scientific, L3000001), and 20 μ l of Lipofectamine P3000 reagent were prepared in 250 μ l of Opti-MEM (Gibco, 11058021). This transfection mix was gently added by dripping onto 1.5×10^6 HEK293T cell monolayer in 2 ml of DMEM containing 10% FBS for 16 hours. Transfection medium was removed and replaced with 2 ml of 10% FBS DMEM. Forty-eight hours posttransfection, lentivirus supernatants were collected and centrifuged at $\times 1000g$ [relative centrifugal forces (RCF)] for 5 min to pellet cell debris and stored at -80°C . Collected lentiviruses were each transduced into U4C *JAK1*^{-/-} cells with single-ISG lentiviruses for 24 hours in the presence of polybrene (8 μ g/ml; Sigma-Aldrich, H9268), before puromycin antibiotic selection (0.4 μ g/ml). Cells remained under selection until all cells in the nontransduced cells were floating or dead. Successful transductants were confirmed for red fluorescent protein expression and stored at -80°C .

Synthetic modRNA synthesis

Production of in vitro transcription template constructs and subsequent RNA synthesis have been described previously (55, 56). All oligonucleotide reagents were synthesized by Integrated DNA Technologies. ORFs were amplified by PCR from plasmids encoding emerald GFP, firefly luciferase, and human ISGs (table S1) or their mouse (table S2) or hamster (table S3) orthologs. Because of the Syrian golden hamster genome reference being incompletely annotated, hamster orthologs were selected by querying the Chinese hamster or the Chinese hamster ovary cell line genome reference using the Ensembl API (v100). For *IFI6*, the closest homolog sequence (*Ifi27l2*) was chosen (table S3). According to the manufacturer's instructions, PCR reactions were performed with KAPA HiFi Hotstart ReadMix PCR Kit (Roche, 07958927001). Splint-mediated ligations were carried out with Ampligase Thermostable DNA Ligase (Epicenter Biotechnologies). Untranslated region (UTR) ligations were conducted in the presence of 200 nM UTR oligos and 100 nM splint oligos. All intermediate PCR and ligation products were purified with QIAquick spin columns (QIAGEN) before further processing. Template PCR amplicons were subcloned with the pcDNA 3.3-TOPO TA cloning kit (Invitrogen). Plasmid inserts were excised by restriction digest and recovered with SizeSelect gels (Invitrogen) before being used to template polyadenylate tail PCRs. RNA was synthesized using the MEGAscript T7 kit (Thermo Fisher Scientific, AMB13345), with 8 to 16 μ g of purified tail PCR product serving as the template for each 400 μ l of reaction. A custom ribonucleoside blend was used, consisting of 3'-O-Me-m7G(5')ppp(5')G cap analog (TriLink Bio-Technologies, N-7003-10) and adenosine triphosphate (ATP), guanosine triphosphate (GTP), and cytosine triphosphate (CTP) (all three from MEGAscript kit), along with N1-methylpseudouridine

(TriLink Biotechnologies, N-1081-100). Final nucleotide concentrations in the reaction mixture were 5 mM for the cap analog, 1.35 mM for GTP, 6.86 mM for ATP and CTP, and 6.91 mM for N1-methylpseudouridine. After synthesis, RNA was treated with Antarctic Phosphatase (New England Biolabs, M0289L and B0289S) for 30 min at 37°C to remove residual 5'-phosphates. RNA was then purified and concentrated using RNA Clean & Concentrator spin columns (Zymo Research, R1019). The resulting modRNA was either eluted and precipitated with 5 M ammonium acetate and resuspended in 10 mM tris-HCl and 1 mM EDTA at 100 ng/μl for in vitro use or 20 to 30 μg/μl for in vivo use or left in water as appropriate.

LNP packaging and transfection

modRNA and Lipofectamine RNAiMAX (Thermo Fisher Scientific, 13778150) transfection agent were each dissolved separately in Opti-MEM (Invitrogen), combined, and then incubated for 15 min at room temperature to generate the transfection mixture. For the 10-ISG collection treatments, modRNA for each ISG was equimolarly pooled before packaging within RNAiMAX lipid reagent. A total of 5 or 0.5 μl of RNAiMAX reagent was used for every microgram of modRNA for in vitro or in vivo transfection in hamsters. In vitro transfection was performed by adding the transfection mixture to cells plated in DMEM with 2% FBS. For in vivo transfection of hamster lungs, the transfection mixture was administered either intranasally by a liquid drip into the nasal cavity or intratracheally by direct instillation. Alternatively, modRNAs administered in mice were packaged in Dlin-MC3-DMA formulated with 1,2-distearoyl-*sn*-glycero-3-phosphocholine (DSPC), cholesterol, 1,2-dimyristoyl-rac-glycerol-methoxypolyethylene glycol (DMG-PEG), and 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP) as previously described (57). Dlin-MC3-DMA: DSPC: cholesterol:DMG-PEG was prepared with a molar ratio of 50:10:38.5:1.5 with 50% DOTAP.

IFN- α 2b cell stimulation and exposure experiments

Control or *ISG15*^{-/-} fibroblasts were stimulated with IFN- α 2b (0 or 1000 IU/ml; Merck, IntronA) in normal growth medium for 12 hours. The IFN- α 2b was then eliminated by thorough washing (3×) with PBS. The cells were allowed to rest for 36 hours in normal medium before infection or being passed off to other downstream assays.

Reverse transcription quantitative polymerase chain reaction

Cell lines were lysed, and RNA was isolated with RNeasy spin columns (QIAGEN, 74104). Reverse transcription was performed with the High-Capacity RT Kit (Applied Biosystems, 4368814). The resulting cDNA was then subjected to qPCR with the TaqMan Master Mix II with UNG (Thermo Fisher Scientific, 4440038), on a Roche LightCycler 480 or Applied Biosystems QuantStudio 6, with the following primers/probes: *18S* (4318839), *MX1* (Hs00895608), *RSAD2* (Hs00369813), *SIGLEC1* (Hs00988063), *IFIT1* (Hs01911452), *BST2* (Hs00171632), *USP18* (Hs00276441), *ISG15* (Hs01921425), *IFI44L* (Hs00915292), and *GAPDH* (Hs01922876). The relative expression of each transcript was normalized relative to *18S* or *GAPDH* by the $\Delta\Delta C_t$ method.

Mass spectrometry

U4Cs were washed in ice-cold PBS and collected in PBS. Cell pellets were lysed in 100 μl of 8 M urea, 50 mM ATP-binding cassette, 150 mM NaCl, and Halt Protease Inhibitors (78429) in Thermo Fisher

Scientific Gold HPLC water (W64). Protein concentration was measured with the Thermo Fisher Scientific Micro BCA kit (23235). A total of 20 μg of lysate was removed from each condition. Protein was then reduced with tris(2-carboxyethyl)phosphine to a final concentration of 2 mM (PG82089). After reduction, proteins were alkylated with iodoacetamide (IAA) to a final concentration of 10 mM (i6125-5g). Excess IAA was quenched with dithiothreitol (DTT) to a final concentration of 10 mM (D5545-1G). Four volumes of 50 mM ammonium bicarbonate were added to dilute the urea to less than 2 M. Proteins were digested with a 1:100 trypsin-to-protein ratio. Digestion took place over 16 hours with end-over-end rotation at room temperature. Samples were acidified to 0.03% trifluoroacetic acid to stop the digestion. Samples were then desalted according to the manufacturer's protocol using Nest Group C18 spin columns (HUM S18R) and dried by vacuum centrifugation.

All samples were analyzed on an Orbitrap Eclipse MS system (Thermo Fisher Scientific) equipped with an Easy-nLC 1200 ultrahigh-pressure liquid chromatography system (Thermo Fisher Scientific) interfaced by a Nanospray Flex nanoelectrospray source. For all analyses, 1 μl of sample containing 0.5 μg of digested peptides was injected onto a C18 reverse phase column (30 cm-by-75 μm inner diameter) packed with Reprosil-Pur 1.9-μm particles. The mobile phase A consisted of 0.1% formic acid (FA), and mobile phase B consisted of 0.1% FA/80% acetonitrile. Peptides were separated by an organic gradient from 5 to 35% mobile phase B over 120 min, followed by an increase to 100% B over 10 min at a flow rate of 300 nl/min. The analytical columns were equilibrated with 3 μl of mobile phase A. Samples from each set of biological replicates were pooled and analyzed with a data-dependent acquisition (DDA) method to build a spectral library. DDA was performed by acquiring a full scan over a mass/charge ratio (*m/z*) range of 375 to 1025 in the Orbitrap at 120,000 resolving power (at 200 *m/z*) with a normalized automatic gain control (AGC) target of 100%, a radio frequency (RF) lens setting of 30%, and an instrument-controlled ion injection time. Dynamic exclusion was set to 30 s with a 10-parts per million exclusion width setting. Peptides with charge states 2 to 6 were selected for tandem mass spectrometry (MS/MS) interrogation using higher-energy collisional dissociation (HCD) with a normalized HCD collision energy of 28%, with 3 s of MS/MS scans per cycle. Individual samples were subsequently analyzed with a data-independent acquisition (DIA) method composed of a full MS scan at 60,000 resolving power over a scan range of 390 to 1010 *m/z*, an instrument-controlled AGC target, an RF lens setting of 30%, and an instrument-controlled maximum injection time, followed by DIA scans using 8 *m/z* isolation windows over 400 to 1000 *m/z* at a normalized HCD collision energy of 28%.

DDA data were searched using Spectronaut to identify peptides and proteins and build a spectral library as previously described (58). Data were searched against the Swiss-Prot portion of the UniProt *Homo sapiens* reference proteome (downloaded on 10 October 2019). False discovery rates (FDRs) were estimated using a decoy database strategy. All data were filtered to achieve an FDR of 0.01 for peptide-spectrum matches, peptide identifications, and protein identifications. Search parameters included a fixed modification for carbamidomethyl cysteine and variable modifications for N-terminal protein acetylation and methionine oxidation. All other search parameters were set to the defaults for the respective algorithms. The MSstats statistical package in R was subsequently applied to estimate log₂ fold changes (FCs), *P* values, and adjusted *P* values (59). All data were normalized by equalizing median intensities, the summary method used was

Tukey's median polish, and the maximum quantile for deciding censored missing values was 0.999. Significantly dysregulated proteins were defined as those that had a fold change value > 2 or < -2 , with a P value of < 0.05 .

Virus propagation and virus titration

A working stock of SARS-CoV-2 (isolate USA-WA1/2020, BEI Resources, NR-52281) was prepared by a low MOI of Vero-E6 cells with SARS-CoV-2 isolate (passage 2) as previously described (60). Infected cultures were maintained in reduced serum DMEM with 2% FBS for 4 days before preparation and storage of virus aliquots at -80°C . Viral titers are reported as plaque-forming units (PFU) of SARS-CoV-2. Propagations and experiments were performed in a biosafety level 3 (BSL-3) conventional biocontainment facility, a dean's CoRE facility, in compliance with the approval of the ISMMS Institutional Biosafety Committee. A/Viet Nam/1203/04 H5N1 (H5N1) virus previously described in (61) was propagated in 10-day-old embryonated chicken eggs at 37°C . Allantoic fluid was harvested after 48 hours, and aliquots were kept at -80°C . Viral titer was determined by plaque assay in Madin-Darby canine kidney cells (MDCK cells) in the presence of $1\text{ }\mu\text{g/ml}$ of tosyl-L-phenylalanine chloromethyl ketone-treated trypsin. VSV and ZIKV were propagated by infecting 10^7 Vero cells at an MOI of 0.01. Supernatants were collected starting at 48 hours postinfection at either 12- or 24-hour intervals until most of the cells had died. Collected supernatants were clarified by filtration through a $0.45\text{-}\mu\text{m}$ filter and stored in aliquots at -80°C .

In vitro viral infections

Cells were infected with VSV-GFP, IAV H5N1, ZIKV (PRVABC59), or SARS-CoV-2 virus at indicated MOIs and durations (detailed in Figs. 2 and 3). For SARS-CoV-2 infections, U4C *JAK1*^{-/-} IFN-I nonresponsive cells were first transduced with human angiotensin-converting enzyme 2 (ACE2) to render them susceptible to infection. ACE2 expression was confirmed by flow cytometry using purified mouse antihuman ACE2 antibody (BioLegend, 342752; 1:200), followed by an allophycocyanin-conjugated goat antimouse immunoglobulin G (IgG) secondary antibody (BioLegend, 405308; 1:200). ACE2⁺ U4Cs were then sorted, expanded, and used for infection assays. After a 1-hour viral adsorption, cells were washed to remove unbound virus and incubated for the remainder of the infection period. At the end of the infection (detailed in Figs. 2 and 3), cells were fixed in 4% paraformaldehyde (PFA) for 20 min at 4°C and stained with 4',6-diamidino-2-phenylindole (DAPI) (Thermo Fisher Scientific, D1306; 1:5000). For fluorescent viruses (VSV-GFP), infected cells were imaged for GFP and DAPI fluorescence. For nonfluorescent viruses (ZIKV, WNV, and SARS-CoV-2), cells were permeabilized and stained with virus-specific antibodies: anti-flavivirus group antigen antibody (Millipore Sigma, MAB10216) for ZIKV and WNV, used at 1:100 dilution, and anti-SARS-CoV-2 nucleoprotein rabbit polyclonal antibody (clone 1C7C7; a gift from T. Moran), used at 1:1000 dilution. After primary antibody incubation, cells were stained with antirabbit IgG Alexa Fluor 488 (Thermo Fisher Scientific, A-11008) secondary antibody at a 1:2000 dilution. Fluorescent images were acquired on a Celigo Imaging Cytometer (Nexcelom) and quantified in CellProfiler (62). DAPI was segmented (global; Minimum Cross-Entropy) to determine total nuclei, and infection was quantified by identifying nuclei overlapping with virus-specific fluorescence signals.

In vivo viral infections

Syrian golden hamsters were anesthetized with ketamine/xylazine before experimental manipulation and euthanized by intracardiac administration of Sleepaway euthanasia solution. Experiments with SARS-CoV-2-infected hamsters were performed in an animal biosafety level 3 (ABSL-3) biocontainment facility at ISMMS. Hamsters were intranasally inoculated with 1×10^5 PFU (titered on Vero-E6 and diluted in $100\text{ }\mu\text{l}$ of PBS) of SARS-CoV-2 virus. Similarly, mice were anesthetized with ketamine/xylazine [ketamine (95 mg/kg) and xylazine (10 mg/kg)] and intranasally inoculated with $30\text{ }\mu\text{l}$ of PBS containing 500 PFU (titered on MDCK cells) of IAV strain A/PR/8/34 (H1N1) (PR8) virus.

In vitro viral titer quantification

TCID₅₀ viral titering was conducted as previously described in (60, 63). Briefly, supernatants were collected from infected cell cultures, centrifuged at $1000g$ (RCF) for 10 min at 4°C to remove cell debris, and serially diluted. A day before infection, adherent target cells were seeded in 96-well plates to reach confluency. Diluted viral supernatants were added to the cells and incubated for 7 days. At end point, cells were fixed with 4% PFA and stained with DAPI. Fluorescent reporters or virus-specific antibodies were used to identify infected cells, as described above. Plates were scanned using the Celigo Imaging Cytometer, and TCID₅₀ values were calculated according to established methods (63).

Plaque assay for quantification of viral titers of tissue specimens

To determine viral titers from infected animal tissues, lungs were first harvested and homogenized in a defined volume of infection medium using a bead mill homogenizer. Homogenates were clarified by centrifugation, and supernatants were used for plaque assays. For the assay, 1×10^6 Vero-E6 cells were seeded in a six-well plate the day before the plaque assay was performed. Briefly, 10-fold serial dilutions were prepared in infection medium for SARS-CoV-2 and inoculated onto a confluent Vero-E6 cell monolayer. After 1-hour adsorption, supernatants were removed, and cell monolayers were overlaid with minimum essential medium (MEM) containing 2% FBS and purified agar (Oxoid) at a final concentration of 0.7%. Cells were then incubated for 3 days at 37°C . Cells were fixed overnight with 4% formaldehyde to inactivate any potential live SARS-CoV-2 virus. The overlay was removed, and cells were washed once with PBS. Plaques were visualized by staining with 1% crystal violet (Sigma-Aldrich, C0775) for 15 min. Viral titers were calculated considering the number of cells per well and expressed as PFU normalized by tissue weight. A similar protocol was followed in MDCK cells to quantify the IAV strain A/PR/8/34 (H1N1) (PR8). Briefly, MDCK cells were seeded in a 12-well plate the day before the plaque assay was performed. Tenfold serial dilutions were performed in infection medium for IAV PR8 and inoculated onto a confluent MDCK cell monolayer. After 1-hour adsorption, supernatants were removed, and cell monolayers were overlaid with MEM containing 0.1% NaHCO_3 , 0.05% diethylaminoethyl-dextran, and purified agar (Oxoid) at a final concentration of 0.6%. Cells were then incubated for 2 days at 37°C in 6% CO_2 . Cells were fixed with 2% formaldehyde solution, followed by staining with 1% crystal violet (Sigma-Aldrich, C0775), and viral titers were expressed as PFU per gram of tissue.

Histological analysis

Hamster lungs were collected and fixed in 10% neutral buffered formalin for 24 hours. Subsequent tissue staining and scoring were performed by iHisto Inc. Briefly, samples were embedded in paraffin and sectioned at 4 μm . Sectioned samples were placed on glass slides, stained with hematoxylin and eosin, and then scanned to create whole slide images. These images were evaluated by a board-certified veterinary pathologist. Study groups and individual animal treatments were not revealed to the pathologist, and slides were evaluated in a blinded manner. Microscopic findings were identified and assigned a severity score of either minimal (1 to 20% tissue affected), mild (21 to 40% of tissue affected), moderate (40 to 60% of tissue affected), marked (61 to 80% of tissue affected), or severe (greater than 80% of tissue affected).

Cell death assay

Fibroblasts were plated at ~80 to 90% confluency in a 96-well plate and kept overnight in growth medium containing 10% FBS. As a positive control for test conditions, 10 μM of shikonin (Santa Cruz Biotechnology, #517-89-5) or 10% DMSO was added to induce cell death. After a 24- or 48-hour incubation, viability was measured using the Deep Blue Cell Viability Kit (BioLegend, 424702). Alternatively, cells were collected and stained with Zombie NIR dye (BioLegend, 423105) or with anti-cleaved caspase-3 antibody (BioLegend, 559341) for quantification by flow cytometry.

Enzyme-linked immunosorbent assay

Supernatants were collected from A549, 2fTGH, and HEK293T cells. Enzyme-linked immunosorbent assays (ELISAs) were performed according to the manufacturer's protocols. ELISAs were used to quantify human IFN- β (R&D Systems, DY814-05PBL, and Assay Science, 41410-2), human IFN- γ (BioLegend, 43104), and human IFN- λ 4 (AFG Bioscience, EK716122, and VWR, 76578-602).

Immunoblots

Cells were lysed in radioimmunoprecipitation assay buffer (Thermo Fisher Scientific, 89900) containing protease/phosphatase inhibitor cocktail (Cell Signaling Technology, 5872) for 20 min. Lysates were centrifuged at 14,000 rpm for 10 min to remove insoluble complexes. Supernatants were boiled with NuPAGE sample buffer (Thermo Fisher Scientific, NP0007) containing 50 mM DTT. The samples were run by gel electrophoresis, and proteins were transferred onto polyvinylidene difluoride membranes. Membranes were blocked with 5% milk and incubated overnight with primary antibodies [rabbit anti-radical S-adenosyl methionine domain-containing 2 (RSAD2): Cell Signaling Technology, clone DST2X, 13996 (1:1000); rabbit polyclonal anti-IFI6/IFI27: Abcam, ab171919; rabbit anti-phospho-Tyr-689-STAT2: Cell Signaling Technology, clone D3P2P, 88410 (1:1000); rabbit anti-phospho-Tyr-701-STAT1: Cell Signaling Technology, clone 58D6, 9167 (1:1000); rabbit anti-USP18: Cell Signaling Technology, clone D4E7, 4813 (1:500); and mouse anti-glyceraldehyde-3-phosphate dehydrogenase (GAPDH): Millipore, clone 6C5, MAB374 (1:10,000)]. Next, membranes were incubated with horseradish peroxidase (HRP)-conjugated secondary antibodies for 1 hour (Goat anti-Rabbit IgG HRP, SouthernBioTech 403005 or Goat anti-Mouse IgG HRP, SouthernBioTech 101005). Pierce SuperSignal West Pico or Femto Chemiluminescent substrate (Thermo Fisher Scientific, 34580/34096) was added to the membranes, and bands were visualized using Amersham ImageQuant 600 (Cytiva).

RNA-seq and relevant analyses

ISG15^{-/-} and control unstimulated and IFN-I-exposed [12-hour stimulation; IFN- α 2b (1000 IU/ml), 24-hour rest] human telomerase reverse transcriptase (hTERT)-immortalized fibroblasts and lung tissues from Syrian golden hamsters treated with PBS or modRNA-LNP formulations on day 4 after SARS-CoV-2 infection were lysed. Next, total RNA was isolated using the Rneasy spin columns (QIAGEN, 74104) and sent for high-throughput RNA-seq to Azenta Life Sciences. FASTQ files were uploaded to BaseSpace (Illumina) and aligned to the human genome build hg19 or the *M. auratus* (MesAur1.0 assembly) reference genome with the RNA-seq alignment module Spliced Transcripts Alignment to a Reference (STAR). The resulting sample-gene expression count matrices were exported and read into the R (v4.0.4) statistical computing platform. Metadata and experimental design were extracted from sample names and used to generate a design matrix for differential gene expression (DGE) analyses. Before DGE analyses, sample-gene matrices were filtered to exclude lowly expressed genes (counts <5 in three or more samples). DGE analyses for fibroblasts were conducted using DESeq2 (v1.30.1) under a paired-sample responsive design (design: ~genotype + IFN-I stimulation). Resulting DGE lists were filtered against a log₂FC of 1.5, and known ISGs (54) were subset from this list. For hamster samples, DGE was performed using DESeq2 (V1.30.1) modeling treatment effects. Viral gene expression was quantified by normalized counts of SARS-CoV-2 transcripts in hamster samples. Variance stabilizing normalization function was applied to the sample-gene count matrices, and differentially responsive genes were visualized using the ComplexHeatmap package (v2.11.1) and ggplot2 (v3.3.5). PCA was performed on hamster samples using variance-stabilized counts to evaluate sample clustering by treatment. Differentially responsive ISGs between conditions were subject to Gene Ontology (GO) analysis using the goana function inside the edgeR package (v3.32.1). Briefly, gene symbols were converted into the Entrez IDs by mapping to the human genome annotation database (org.Hs.eg.db; v3.12.1) and fed into the goana function. To mitigate *P* value exaggeration and control for background specificity, the gene universe was explicitly set only to include genes detected and used for DGE. Resulting GO terms were filtered against a $P < 1 \times 10^{-12}$ and subset to only include biological process terms. Passing GO terms were visualized using ggplot2 (v3.3.5).

Statistical analysis

Individual-level data are presented in data file S1. All statistical analyses were performed using GraphPad Prism (GraphPad Software Inc.) or R. For comparisons between two groups, data were analyzed using unpaired or paired two-sided *t* tests, with Welch's correction applied when variance equality was not assumed. If data did not meet normality assumptions, as assessed by the Shapiro-Wilk test, then a nonparametric Mann-Whitney test was used. For comparisons involving more than two groups, a one-way analysis of variance (ANOVA) was used, followed by Dunnett's or Tukey's post hoc tests, depending on whether comparisons were made to a single control or across all groups. When assumptions of normality or homogeneity of variance (evaluated using Levene's test) were not met, data were either log-transformed or analyzed using a nonparametric Kruskal-Wallis test, followed by Dunn's multiple-comparison test, as indicated in the figure legends.

In longitudinal animal studies, particularly when animal drop-out occurred because of humane end points, a mixed-effects model

using restricted maximum likelihood with Geisser-Greenhouse correction was applied to assess group and time interactions. Post hoc comparisons were performed using Dunnett's test to compare treated groups with the control at each time point. For RNA-seq data, differential expression was calculated using the Wald test in DESeq2, with *P* values adjusted for multiple testing using the Benjamini-Hochberg method to control the FDR.

Survival data were visualized using Kaplan-Meier survival curves. Because all animals in these experiments reached the pre-defined humane end point, no censoring occurred. Group comparisons were performed using a two-sided log-rank (Mantel-Cox) test. To confirm the validity of the survival analysis, proportional hazard assumptions were evaluated by visually inspecting log-minus-log survival plots and testing scaled Schoenfeld residuals. No violations of the proportional hazard assumption were detected.

The threshold for statistical significance was set at $\alpha = 0.05$. All hypothesis testing was two sided unless otherwise specified. Where reported, confidence intervals were based on SDs (mean $\pm$ SD). The validity of each statistical approach was confirmed by verifying that the underlying assumptions were met: normality of residuals (Shapiro-Wilk test and quantile-quantile plots), homogeneity of variance (Levene's test), and suitability of model fit for longitudinal or survival data. Where assumptions could not be met or corrected by transformation, appropriate nonparametric alternatives were used.

Supplementary Materials

The PDF file includes:

Figs. S1 to S10

Tables S1 to S3

Other Supplementary Material for this manuscript includes the following:

Data file S1

MDAR Reproducibility Checklist

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reports ownership in Lab11 Therapeutics. A patent (D.B. and J.T., "Gene combination as a broad-spectrum antiviral," 9 June 2022, application no. 63/208,827) covering the usage of the 10-ISG collection as a broad-spectrum antiviral has been filed by the Icahn School of Medicine at Mount Sinai. B.R.R. is on the scientific advisory board of Dispatch Biotherapeutics. The A.G.-S. laboratory has received research support from Pfizer, Senhwa Biosciences, Kenall Manufacturing, Avimex, Johnson & Johnson, Dynavax, 7 Hills Pharma, PharmaMar, ImmunityBio, Accurius, nanoComposix, Hexamer, N-fold LLC, Model Medicines, Atea Pharmaceuticals, Applied Biological Laboratories, and Merck, outside of the reported work. A.G.-S. has consulting agreements for the following companies involving cash or stock: Vivaldi Biosciences, ContraFect, 7 Hills Pharma, Avimex, Vaxalto, Pagoda, Accurius, Esperovax, Farmak, Applied Biological Laboratories, PharmaMar, Paratus, CureLab Oncology, CureLab Veterinary, Synairgen, and Pfizer, outside of the reported work. A.G.-S. has been an invited speaker in meeting events organized by Seqirus, Janssen, and AstraZeneca. A.G.-S. is the inventor of patents and patent applications on antivirals and vaccines for the treatment and prevention of

virus infections and cancer, owned by the Icahn School of Medicine at Mount Sinai, New York, outside of the reported work. B.R.R. is a member (compensated) of the Scientific Advisory Board for Dispatch Biotherapeutics. **Data and materials availability:** All data associated with this study are present in the paper or the Supplementary Materials. RNA-seq data (raw and processed), along with analysis scripts and visualization code reported here, can be found deposited in Gene Expression Omnibus under accession numbers GSE301682 and GSE301683. Requests for resources, reagents, and additional information should be directed to the lead contact, D.B. (db3700@cumc.columbia.edu). Requests involving constructs or noncommercial reagents will require completion of a material transfer agreement.

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An mRNA-based broad-spectrum antiviral inspired by ISG15 deficiency protects against viral infections in vitro and in vivo

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Editor's summary

The type I interferon (IFN-I) response is a conserved cascade of signaling and gene expression that, among other functions, confers protection of cells from viral infection. After resolution of infection, the response is tamped down by regulators such as IFN-I-stimulated gene 15 (ISG15). Cells from individuals lacking ISG15 are able to control viral infections in vitro as a consequence of maintaining a low-grade IFN-I response. Inspired by this observation, Akalu *et al.* identified a set of 10 ISGs that mimicked what is observed in cells from individuals lacking ISG15, with the idea that these 10 ISGs could serve as a broad-spectrum antiviral. The authors found that the 10 ISGs enabled control of multiple viral infections in vitro and lessened disease severity of SARS-CoV-2 when prophylactically administered as mRNAs to mice. Although limited delivery of the mRNAs may have restricted efficacy, as discussed by the authors, these data lay the foundation for development of a broad-spectrum antiviral prophylactic. —Courtney Malo

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